you just memorize this pattern: Switch the derivative from one factor of the integrand to the other at the cost of a minus sign.

8.1.4 Linear Operators

Bras and kets are half the story in quantum mechanics; the other half is the concept of linear operators and, in particular, Hermitian operators. This raises two questions:

- What is meant by a linear operator on a space of functions?
- What is the condition for a linear operator to be Hermitian?

The concept of a linear operator is simple enough: it's a machine that acts on a function and gives another function. When it acts on the sum of two functions, it gives the sum of the individual results. When it acts on a complex numerical multiple of a function, it gives the same multiple of the original result. In other words, it is (surprise!) linear.

Let's look at some examples. One simple operation we can perform on a function $\psi(x)$ is to multiply it by x. That gives a new function $x\psi(x)$, and you can easily check that the action is linear. We'll represent the "multiply by x" operator with the symbol **X**. By definition, then,

$$\mathbf{X}\ \psi(x) \ = \ x\psi(x). \qquad (8.5)$$

Here's another example. Define **D** to be the differentiation operator:

$$\mathbf{D}\ \psi(x) \ = \ \frac{d\psi(x)}{dx}. \qquad (8.6)$$

Exercise 8.1: Prove that **X** and **D** are linear operators.